

William D. Kistler, PhD

National Institute of Neurological Disorders and Stroke, National Institutes of Health
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[Personal](#) — [GitHub](#) — [LinkedIn](#)

Profile

Computational neuroscientist trained in motor skill learning, movement quantification, and machine-learning analyses for behavioural and clinical data. My doctoral research examined how multi-effector kinematic motifs emerge and change during motor learning, linking fine-grained movement data to trial-level changes in skilled behaviour. Additionally, I have over five years experience in clinical research, virtual reality, digital health, and translational software development for rehabilitation and cognitive support.

Education

PhD in Neuroscience

National Institutes of Health and UCL Queen Square Institute of Neurology

Thesis: *Examining the Time-Course of Multi-Effector Synergy in Motor Sequence Learning*

OxML 2024

University of Oxford

Lectures and practicals on ML in health and biology, representation learning, and generative AI.

BS, Psychology, *cum laude*

University of Maryland, College Park

Thesis: *Longitudinal Socio-Cognitive Development in Children at Play*

Research Appointments

PhD Fellow, National Institute of Neurological Disorders and Stroke, NIH, Bethesda, MD 2020–2025

Doctoral research on motor sequence learning, kinematic synergies, and computational analysis of high-dimensional behavioural data. Developed pipelines for remote behavioural testing, pose estimation, dimensionality reduction, unsupervised clustering, and distributional comparison of trial-level motor patterns.

Lab Manager, Immersive Simulation Program, NIH, Bethesda, MD 2015–2019

Managed virtual-reality and behavioural-science studies in health communication, clinical decision-making, nutrition, and neurodevelopmental research. Supported experimental design, data collection, analysis, and trainee supervision.

Research Volunteer, Functional Applied Biomechanics Lab, NIH, Bethesda, MD 2015–2019

Developed research on motor rehabilitation in virtual reality for children with cerebral palsy.

Research Technician, Psychophysics Lab, American University, Washington, DC 2013–2014

Contributed to psychophysics experiments examining motion, colour, visual perception, and material perception.

Research Assistant, Center for Behavioural Neuroscience, American University, Washington, DC 2011–2012

Contributed to psychophysics experiments examining motion, colour, visual perception, and material perception.

Post-Baccalaureate IRTA Fellow, National Institute on Aging, NIH, Rockville, MD 2007–2009

Conducted research as part of the NIH Intramural Research Training Award programme.

Research Assistant, Child Development Laboratory, University of Maryland, College Park, MD 2005–2007

Supported longitudinal research on child development and social cognition.

Research Assistant, Motor Development Group, University of Maryland, College Park, MD 2005–2007

Supported research on motor development and behavioural assessment.

Publications

Journal Articles

Kistler, W., Fakhreddine, R., Rodriguez, G. R., Hayward, M., Buch, E. R., Bestmann, S., & Cohen, L. G. (under revision). Early Skill Learning Is Shaped by the Offline Emergence of Expert Synergies. Manuscript under revision at *Current Biology*.

- Srivastava, A., Marabelli, M., et al. (2025). The Present and Future of AI: Ethical issues and research opportunities. *Communications of the Association for Information Systems*. DOI
- Yeo, Y., Peng, Y., et al. (2024). Evaluating for evidence of sociodemographic bias in conversational AI for mental health support. *Cyberpsychology, Behavior*. DOI
- Kistler, W.***, Mangalmurti, A.*, Quarrie, B., Sharp, W., Persky, S., Shaw, P. (2020). Using virtual reality to define the mechanisms that link symptoms to cognitive deficits in ADHD. *Scientific Reports*. DOI
- Yaremych, H. E., **Kistler, W.**, Trivedi, N., Persky, S. (2019). Path tortuosity in virtual reality: A novel approach for quantifying behavioral process in a food choice context. *Cyberpsychology, Behavior, and Social Networking*. DOI
- Persky, S., Ferrer, R., Klein, W., Goldring, M. R., Cohen, R., **Kistler, W.**, Yaremych, H. E., Bouhlal, S. (2018). Effects of fruit and vegetable feeding messages on mothers and fathers. *Annals of Behavioral Medicine*. DOI
- Persky, S., Goldring, M., Turner, S., Cohen, R., **Kistler, W.** (2018). Validity of assessing child feeding with virtual reality. *Appetite*. DOI
- Persky, S., **Kistler, W.**, Klein, W., Ferrer, R. (2018). Internet versus virtual reality settings for genomics information provision. *Cyberpsychology, Behavior, and Social Networking*. DOI

Journal Abstracts

- Xiao, B., **Kistler, W.** (2015). Perceptual dimensions of material properties of fabrics in dynamic scenes. *Journal of Vision*, 15(12), 938. DOI

Book Chapters

- Kistler, W.** (2016). *Color Wagon-Wheel Illusion*. In *The Oxford Compendium of Visual Illusions*. Oxford University Press.

Patents

- Kistler, W.** (2017). A composite ellipsoidal device facilitating one or more manual therapies and the control of a computer monitor. Filed November 20, 2017. Patent pending.

Honours, Awards & Distinctions

G.R.E.A.T. Award , National Human Genome Research Institute, NIH	2018
Distinguished Achievement Award , National Human Genome Research Institute, NIH	2016
Graduate Research Fellowship Award , American University	2012
Graduate Division Travel Award , American University	2012
Dean's Outstanding Achievement Award , American University	2012
2nd Place , Best Illusion of the Year Contest, Vision Science Society	2012
Outstanding Scholarship at the Graduate Level Award , American University	2011
Cum Laude , University of Maryland	2007

Technical Expertise

Computational neuroscience: Motor learning, kinematic synergies, movement quantification, behavioural modelling, clinical digital phenotyping.

Machine learning: Representation learning, dimensionality reduction, unsupervised clustering, classification, regression, cross-validation.

Statistics: Permutation testing, Jensen–Shannon divergence, distributional comparison, multiple-comparison correction, empirical p-value estimation.

Signal processing and time series: Filtering, wavelet analysis, spectral analysis, artefact detection, feature extraction from high-noise behavioural data.

Programming and tools: Python, NumPy, Pandas, scikit-learn, PyTorch, Git, Bash, C, Cython.

Research systems: Remote behavioural testing, pose estimation workflows, reproducible analysis pipelines, clinician-facing data summaries.

Teaching & Mentorship

Teaching

Guest Lecturer , Queen Square Movement Disorders Short Course, UCL Queen Square IoN	Fall 2024
Guest Lecturer , Upper Limb Neurorehabilitation Course, UCL Queen Square IoN	Summer 2024
Teaching Assistant , Cognitive Neuroscience, American University	Spring 2012
Teaching Assistant , Developmental Biopsychology, University of Maryland	Spring 2007

Mentorship

Mentor , UCL Institute of Neurology Academic Mentoring Scheme	Summer 2024
Mentored Sarah Imdad, UCL IoN PhD student, on data quality in ML for science.	
Mentor , NIH Summer Internship Program in Biomedical Research	Summer 2019
Mentored Sara Flash on machine learning approaches for human behaviour.	
Mentor , NIH Summer Internship Program in Biomedical Research	Summer 2017
Mentored Benjamin Shapiro on computational biomechanics.	

Presentations

Oral Presentations

- Kistler, W.** (2026). Mechanisms Underlying the Formation of Expert Synergies. National Institutes of Health, Clinical Center Grand Rounds, Bethesda, MD.
- Kistler, W.** (2022). At-home patient assessments and the synthetic data problem. National Institute of Mental Health and University College London Research Seminar, London, UK.
- Kistler, W.** (2020). Novel collection methods for behavioural data in the home and machine learning assisted analyses. National Institute of Mental Health and University College London Research Seminar, London, UK.
- Kistler, W., Xiao, B.** (2015). Perceptual dimensions of material properties of fabrics in dynamic scenes. Vision Science Society, St. Pete Beach, FL.
- Kistler, W., Shapiro, A., Rose-Henig, A.** (2014). Separating motion responses in human vision. European Conference on Visual Perception, Belgrade, Serbia.
- Kistler, W., Engel, L., Stogner, M.** (2012). Make believe: how media is made to matter. American University, Washington, DC.

Poster Presentations

- Kistler, W., Fakhreddine, R., Hayward, M., Rodriguez, G., Buch, E., Bestmann, S., Cohen, L. G.** (2022). Kinematic synergies rapidly update to support procedural skill learning. Society for Neuroscience, San Diego, CA.
- Kistler, W., Fakhreddine, R., Hayward, M., Rodriguez, G., Buch, E., Bestmann, S., Cohen, L. G.** (2022). At-home characterization of kinematic synergies supporting procedural skill learning. Neural Control of Movement, Dublin, Ireland.
- Kistler, W., Goldsmith, D., Brennan, P.** (2019). Virtual Reality: Exploring methods to improve dietary choices. American Medical Informatics Association.
- Persky, S., **Kistler, W.** (2017). Virtual reality in healthcare communication research and practice. International Conference on Communication in Healthcare, Baltimore, MD.
- Kistler, W., Persky, S., Bouhlal, S., Sypher, A.** (2015). The Immersive Virtual Environment Testing Area: A virtual tour. National Institutes of Health, National Human Genome Research Institute, Social Behavioral Research Branch Quadrennial Review, Bethesda, MD.
- Kistler, W., Shapiro, A.** (2014). Facial-feature changes are hard to track in the color wagon-wheel illusion. Vision Science Society, St. Pete Beach, FL.
- Kistler, W., Shapiro, A., Rose-Henig, A.** (2014). Hypocycloid variations and apparent motion. Vision Science Society, St. Pete Beach, FL.
- Kistler, W., Xiao, B.** (2014). Perceptual dimensions of material properties in dynamic scenes. European

Conference on Visual Perception, Belgrade, Serbia.

Kistler, W. (2013). *Im/Material: Human vision and the digital concern*. Society for Cinema and Media Studies, Chicago, IL.

Shapiro, A., **Kistler, W.**, Flynn, O. (2013). Contrast-contrast asynchrony effects in human vision. Vision Science Society, Naples, FL.

Shapiro, A., **Kistler, W.**, Rose-Henig, A. (2012). The color wagon wheel. Vision Science Society, Naples, FL.

Research Software & Projects

Lucy: Dementia-Friendly AI Companion

[GitHub](#)

Python, PyTorch, Flask, zero-shot classification, encryption, role-based access control

Developed a prototype conversational system for cognitive support and longitudinal language monitoring in dementia care. The project combines supportive dialogue, response filtering, and security-oriented design for handling sensitive patient-facing interactions.

Juno: Machine-Learning-Enabled Patient Scheduler

[GitHub](#)

Python, scikit-learn, Pandas, Flask, PyTest

Developed a prototype scheduling system for rehabilitation wards that groups patients by impairment similarity and provides clinician-facing explanations for schedule recommendations. The project explores how machine-learning systems could support ward-level rehabilitation planning while preserving clinician oversight.

Stella: Self-Supervised Learning for Clinical Phenotyping of Pose Data

[GitHub](#)

Python, SimCLR, UMAP, HDBSCAN, 1D CNN, LSTM

Developed a self-supervised pipeline for clinical phenotyping from pose data. The project uses temporal representation learning with dimensionality reduction and unsupervised clustering to support movement-based assessment of motor impairment.

ColourS: Therapeutic Programs for Eye-Hand Coordination

[GitHub](#)

C, SDL2, OpenGL

Developed therapeutic programs for assessment and training of eye-hand coordination, including real-time feedback and clinically relevant data logging for motor-skill applications.

Service, Outreach & Volunteering

Volunteer, National Head Injuries Association / Headway

2022–2024

Developed community-based outreach questionnaires and programmes for individuals recovering from brain injury.

Interactive Technologies Consultant, The Children's Inn at NIH

2017–2020

Worked with staff to deploy immersive technologies for children and families residing at the Inn.

STEM Outreach Speaker, National Institutes of Health

2016–2019

Conducted neuroscience and behavioural-science workshops for under-represented students.

Tutor, Code.org

2016

Mentored students applying to university at the intersection of coding, statistics, and hypothesis testing.

Welcome Desk Aide, The Children's Inn at NIH

2016–2020

Welcomed families from thirteen countries while they were residing at NIH for research participation.

Event Assistant, Cerebral Palsy Foundation

2014

Assisted staff before and after fundraising events supporting mobility technologies for children with cerebral palsy.

Professional Memberships

Neural Information Processing Systems

2022–Present

Neural Control of Movement

2019–Present

Society for Neuroscience

2013–Present

Institute of Electrical and Electronics Engineers

2013–Present

Vision Science Society

2011–2015